

A-level

# ENVIRONMENTAL SCIENCE

## Paper 2 - Generated practice paper Set 01

### Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

### Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 01.1     | <ul style="list-style-type: none"> <li>• EDGE: Evolutionarily Distinct and Globally Endangered - few close relatives, often the last species of their genus / family (Bactrian camel, pygmy hippo, northern bald ibis); loss would remove a unique branch of evolution / genes</li> <li>• Endemic: found in only one area (often islands - Galapagos, Madagascar, Seychelles: Aldabra giant tortoise, red ruffed lemur); if lost locally there is no population elsewhere to recolonise</li> <li>• Keystone: disproportionately important for the structure of the community relative to its abundance - predation of dominant species, seed dispersal, habitat creation (jaguar, forest elephants keeping paths and water holes open, wolves, beavers); losing them causes loss of many other species</li> <li>• Flagship: high public profile (tiger, panda, elephant, orang-utan) raising funds and support that also protects the whole habitat and less charismatic species (fungi, bats, moths)</li> </ul> | 8     |
| 01.2     | <ul style="list-style-type: none"> <li>• A fragmented population may consist of isolated groups each too small to be viable (resources, inbreeding) even if total numbers are large</li> <li>• A species confined to a single area (eg bonobos in one country) is vulnerable to any local change / conflict, unlike chimpanzees spread across many countries</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 28-29; AQA spec 3.1.2.3.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |
| 02.1     | <ul style="list-style-type: none"> <li>• All greenhouse gases expressed as CO<sub>2</sub> equivalent (allowing for different warming strengths)</li> <li>• Direct emissions from fuel combustion plus indirect: manufacture of components, maintenance items, final disposal (life-cycle analysis)</li> <li>• A fair share of the infrastructure: roads, car parks, signals, street lighting</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 3     |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 02.2     | <ul style="list-style-type: none"> <li>Compares the populations of over 18 000 populations of 3700 vertebrate species (terrestrial, freshwater, marine) with 1970 levels</li> <li>Published in the Living Planet Report as a measure of the state of the planet's ecosystems / biodiversity loss over time</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                      | 2     |
| 02.3     | <ul style="list-style-type: none"> <li>Hardy, disease-resistant Boran cattle; regular tick dipping reduces ticks for wild herbivores too</li> <li>Mobile bomas (night enclosures) protect cattle from lions and leave dung that enriches soil and plant growth</li> <li>Electric fences keep game in and poachers out; biological corridors let animals (except rhinos) move to other reserves; problem elephants relocated</li> <li>Monitoring: radio-collared lions, drones, camera traps, track surveys; endangered species (Grevy's zebra, northern white rhino) protected</li> <li>Community links: ecotourism income funds health, education and water projects; local employment; cattle breeding help for local farmers</li> </ul> | 5     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 380-383; AQA spec 3.6.4.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       |
| 03.1     | <ul style="list-style-type: none"> <li>Environmental DNA (eDNA) analysis</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1     |
| 03.2     | <ul style="list-style-type: none"> <li>Transmitters are too heavy for small birds / their batteries would not last for a long-term study</li> <li>Geolocator tags are small and only record time and light levels; latitude and longitude for each day are worked out when the bird is recaptured and the data downloaded</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                       | 2     |
| 03.3     | <ul style="list-style-type: none"> <li>Each separate (regional) population has its own unique gene pool</li> <li>DNA from seized ivory can be compared with the database to identify where the elephant came from, so poaching hotspots / trade routes can be targeted</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-405; AQA spec 3.1.2.3.6.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |
| 04.1     | <ul style="list-style-type: none"> <li>By-catch = <math>5.9 + 1.9 = 7.8</math> t; total = 8.4 t</li> <li><math>7.8 / 8.4 \times 100 = 92.9\%</math> (accept 93%)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2     |
| 04.2     | <ul style="list-style-type: none"> <li>The nets have a very small mesh size so few non-target animals can escape</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1     |
| 04.3     | <ul style="list-style-type: none"> <li>They breed at an old age (skate about 11 years; Greenland shark 100-150 years) and produce few young</li> <li>They may reach a catchable size before they are sexually mature, so fishing can remove the whole breeding population before it has bred</li> <li>Populations recover slowly, unlike r-selected species (herring, mackerel) which breed young and produce many young; they are caught as by-catch in demersal trawls so fishing continues even though they have no commercial value</li> </ul>                                                                                                                                                                                         | 3     |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 04.4     | <p><b>Levels of response</b></p> <p><b>Level 3 (7-9 marks):</b> Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.</p> <p><b>Level 2 (4-6 marks):</b> Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.</p> <p><b>Level 1 (1-3 marks):</b> Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (credit other relevant, accurate points)</p> <ul style="list-style-type: none"> <li>• Impacts: population decline of k-selected species, by-catch (immature target fish, over-quota fish, non-commercial species; drift nets - whales, dolphins, turtles; longlines - albatrosses; pair trawls - porpoises), ghost fishing, seabed damage by demersal trawls, coral and seagrass damage, dynamite fishing, food-web effects</li> <li>• Catch quotas: limit weight landed; work well for single-species pelagic shoals but in mixed fisheries over-quota fish are discarded dead so the quota fails to protect them</li> <li>• Gear design: mesh size and square-mesh escape panels let small fish escape (diagonal mesh closes under drag); turtle exclusion devices; acoustic pingers for dolphins; curved hooks, decoys, sinkers and night setting to protect albatrosses; biodegradable / radio-tracked traps to reduce ghost fishing</li> <li>• Restricted effort: limits on boat size, engine power and days at sea; decommissioning; restricted methods - drift-net bans, demersal trawling bans on sensitive seabeds (Rockall Bank, Hatton Bank)</li> <li>• Evaluation: enforcement at sea is difficult; some methods reduce catch or raise costs; quotas need reliable stock data; effectiveness depends on species (single vs mixed shoals) and on international cooperation as fish migrate; as local stocks decline fishing moves to more distant waters</li> <li>• Justified conclusion weighing effectiveness of different methods</li> </ul> | 9     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 341-345; AQA spec 3.5.2.2.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |
| 05.1     | <ul style="list-style-type: none"> <li>• Success: Cactoblastis moth introduced from South America controlled prickly pear cactus in Australia; Encarsia parasitic wasps control greenhouse whitefly</li> <li>• Problems: cane toads introduced to Australia to control sugar cane beetles did not control them but ate many other species and are toxic to predators; rosy wolfsnail introduced to Polynesia to control African land snails ate endemic Partulid snails (48 species extinct); harlequin ladybirds introduced to Europe eat native ladybirds</li> <li>• Lesson: control agents should be specialist feeders on the pest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 4     |
| 05.2     | <ul style="list-style-type: none"> <li>• Release an artificial sex scent attracting males (or females) of the pest species</li> <li>• Used to monitor when the pest arrives so pesticide is timed correctly, or (in enclosed greenhouses) to trap enough of one sex to prevent breeding</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2     |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                   | Marks |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 05.3     | <ul style="list-style-type: none"> <li>• Regular low doses kill only the most sensitive bacteria so resistant strains survive and multiply</li> <li>• Agricultural use exceeds medical use, so resistance spreads</li> <li>• Resistant zoonotic bacteria (E. coli, Salmonella, Campylobacter) can transfer to humans and are harder to treat; hence reduced use of antibiotics is a sustainability goal</li> </ul> | 3     |
| 05.4     | <ul style="list-style-type: none"> <li>• Grow nectar-rich flowering plants alongside crops; restrict pesticides harmful to bees / spray when not in flower; introduce bee hives</li> </ul>                                                                                                                                                                                                                         | 1     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 316-320; AQA spec 3.5.1.1.                                                                                                                                                                                                                                                                                            |       |
| 06.1     | <ul style="list-style-type: none"> <li>• About 10% of land is already cultivated and 25% grazed</li> <li>• Most remaining land is too dry, wet, cold, hot or steep, is covered in ice, or has no soil</li> </ul>                                                                                                                                                                                                   | 2     |
| 06.2     | <ul style="list-style-type: none"> <li>• Land becomes too expensive for local people to rent / is used to grow export crops for more affluent societies</li> <li>• So people with less money lose access to the land and food they used to produce</li> </ul>                                                                                                                                                      | 2     |
| 06.3     | <ul style="list-style-type: none"> <li>• It uses natural processes to supply nutrients (legumes, recycled organic matter, soil biota), control pests and provide ecological services</li> <li>• So that pesticides and artificial fertilisers become unnecessary rather than just being banned</li> </ul>                                                                                                          | 2     |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 06.4     | <p><b>Levels of response</b></p> <p><b>Level 3 (7-9 marks):</b> Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.</p> <p><b>Level 2 (4-6 marks):</b> Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.</p> <p><b>Level 1 (1-3 marks):</b> Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (credit other relevant, accurate points)</p> <ul style="list-style-type: none"> <li>• Unsustainable features: reliance on chemical pesticides (resistance, non-target impacts, bans), non-renewable rock phosphate, fossil-fuel-intensive nitrate manufacture, over-exploitation of rivers / groundwater, salinisation, loss of soil organic matter, greenhouse gases from fuel, rice and livestock, loss of species providing ecosystem services</li> <li>• Pest control: cultural methods (weeding, mulching, crop rotation, barrier crops, culling), biological control, predator habitats, polyculture / companion crops, integrated control, reduced antibiotic use</li> <li>• Nutrients: recycling organic matter, crop rotation, cultivation of legumes, conservation of soil biota, reduced artificial fertiliser especially nitrate, low tillage</li> <li>• Water: low water-use crops, reservoirs and aquifer recharge, drip irrigation rather than sprays</li> <li>• Soil and carbon: low-tillage farming to maintain organic matter, reduced machinery, renewable energy, drought-tolerant rice varieties so fields are drained earlier, high-carbohydrate ground feed for cattle reduces methane</li> <li>• Biodiversity: conserve habitats of crop wild relatives, seed banks, retain hedgerows, ditches, ponds, woodland; agri-environment schemes</li> <li>• Evaluation: growing population (perhaps 12 billion) needs more food; lower trophic level diets; obstacles - cost, need for support, gaining acceptance of new techniques; learning from past land degradation (fertile crescent, US mid-west)</li> <li>• Judgement: no single strategy sufficient; combination of ecological principles and new technology; justified conclusion</li> </ul> | 9     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |
| 07.1     | • B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1     |
| 07.2     | <p><i>Any 4 from:</i></p> <ul style="list-style-type: none"> <li>• Deforestation</li> <li>• Ploughing of grassland / expansion of farmland</li> <li>• Reservoir creation / flooding</li> <li>• Mineral extraction, especially open-cast mining</li> <li>• Urban expansion</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26; AQA spec 3.1.2.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>08.1</b> | <i>Any 2 from:</i> <ul style="list-style-type: none"> <li>• Potato bean (N America): high protein content</li> <li>• Morama bean (southern Africa): protein content similar to soya, drought resistant</li> <li>• Yeheb tree (Somalia): edible nuts, drought resistant, grows on poor soils</li> <li>• Perennial maize (Mexico) / Kernza: no need to buy seed each year, less ploughing so less soil erosion</li> <li>• American bison: graze grassland too poor for cattle</li> <li>• Common eland: browse and graze so can be kept with cattle without severe competition</li> </ul> | 2     |
| <b>08.2</b> | <ul style="list-style-type: none"> <li>• Indigenous species are better adapted to local conditions such as climate, pests / diseases and soil</li> <li>• So they have higher survival rates / productivity is not reduced, as it is when introduced species are kept in areas to which they are not adapted</li> </ul>                                                                                                                                                                                                                                                                 | 2     |
| <b>08.3</b> | <ul style="list-style-type: none"> <li>• They grow in soils with a high salt content, so their genes could be used in breeding programmes to increase the salt tolerance of crops grown on soils salinised by irrigation</li> </ul>                                                                                                                                                                                                                                                                                                                                                    | 1     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 11; AQA spec 3.1.2.1.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |
| <b>09.1</b> | <i>Any 2 from:</i> <ul style="list-style-type: none"> <li>• Hydrological cycle</li> <li>• Carbon cycle</li> <li>• Nitrogen cycle</li> <li>• Atmospheric circulation</li> <li>• Thermohaline circulation</li> <li>• Photosynthesis</li> </ul>                                                                                                                                                                                                                                                                                                                                           | 2     |
| <b>09.2</b> | <ul style="list-style-type: none"> <li>• Enzymes reduce the activation energy of reactions</li> <li>• So reactions such as photosynthesis, decomposition and nitrogen fixation do not need high temperatures</li> </ul>                                                                                                                                                                                                                                                                                                                                                                | 2     |
| <b>09.3</b> | <i>Any 1 from:</i> <ul style="list-style-type: none"> <li>• Haber process (nitrogen fixation)</li> <li>• Incineration of wastes (decomposition)</li> <li>• Many manufacturing processes</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                     | 1     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |
| <b>10.1</b> | <i>Any 4 from:</i> <ul style="list-style-type: none"> <li>• Zooxanthellae are photosynthetic algae living inside the polyp cells</li> <li>• The algae absorb CO<sub>2</sub> and excretory wastes from the polyp for photosynthesis and growth (efficient nutrient recycling)</li> <li>• They supply the polyp with oxygen and glucose - over 95% of its food energy</li> <li>• In return the algae gain a protected habitat and nutrients they could not absorb from the nutrient-deficient water</li> <li>• Polyps also feed on plankton captured by cilia and nematocysts</li> </ul> | 4     |
| <b>10.2</b> | <ul style="list-style-type: none"> <li>• Corals deposit a skeleton of calcium carbonate (calcite)</li> <li>• Each carbonate molecule requires the absorption of a molecule of carbon dioxide, so reefs are a carbon sink</li> </ul>                                                                                                                                                                                                                                                                                                                                                    | 2     |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 10.3     | <p><b>Levels of response</b></p> <p><b>Level 3 (7-9 marks):</b> Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.</p> <p><b>Level 2 (4-6 marks):</b> Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.</p> <p><b>Level 1 (1-3 marks):</b> Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (credit other relevant, accurate points)</p> <ul style="list-style-type: none"> <li>• Importance: cover 1% of ocean area but support over a quarter of marine species; fisheries (fish, lobsters, conch); medicines (AZT, Ara-C, dolastatin, secosteroids, bryostatin, trabectedin, bone grafts); climate control as carbon sink; erosion protection of low-lying islands; tourism income</li> <li>• Threats: physical damage (anchors, divers, litter, coral extraction), collection of ornaments (CITES Appendix II), sedimentation from rivers and coastal development, pollution (oil, nutrients causing algal shading, sewage, pesticides, sunscreen), ocean acidification, bleaching from warming, overfishing (giant triton / crown of thorns starfish), introduced species (red lionfish), loss of mangroves and seagrass nurseries</li> <li>• Conservation: marine protected areas / national parks (Great Barrier Reef, Belize, Chagos), mooring buoys, diver education, No Take Zones which can increase catches elsewhere, sustainable ecotourism and fishing, protection of mangroves and seagrass, erosion and pollution control, planning control of ports and resorts</li> <li>• Fishing controls: closed seasons, minimum and maximum catchable sizes, bans on dynamite, trawling and small-mesh nets</li> <li>• Evaluation: local measures cannot control global threats (warming, acidification) which need international action; enforcement in remote areas is difficult; conflicts with local livelihoods; large vs small NTZs</li> <li>• Justified conclusion</li> </ul> | 9     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 11.1     | <p><b>Agriculture involves the control of abiotic and biotic factors to maximise food production. Discuss how this control can be achieved and evaluate its environmental impacts.</b></p> <p><b>Levels of response</b></p> <p><b>Level 5 (21-25 marks):</b> Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.</p> <p><b>Level 4 (16-20 marks):</b> Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.</p> <p><b>Level 3 (11-15 marks):</b> Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.</p> <p><b>Level 2 (6-10 marks):</b> Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.</p> <p><b>Level 1 (1-5 marks):</b> Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (students are not expected to cover all of these; credit other relevant material)</p> <ul style="list-style-type: none"> <li>• Species selection: market, range of tolerance, controllability</li> <li>• Abiotic control: temperature, light, water, soil fertility / nutrients, aeration, salinity, pH, CO<sub>2</sub>, topography, relief, wind - methods and examples</li> <li>• Biotic control: pest groups, cultural control, biological control (successes and failures), sterile males, pheromones, genetic resistance, pesticides (properties), antibiotics, pollinators, soil biota</li> <li>• Manipulation of food species: population density, monocultures, selective breeding, asexual reproduction, cross-breeding, AI / embryo transfer, GM; energetics - intensive vs extensive, energy subsidies, ratios, FCR, food chain losses</li> <li>• Impacts: habitat loss / change, reduced biodiversity, introduced species, pesticide and nutrient pollution, GHGs, hydrological changes, soil erosion; positive: plagioclimax habitats</li> <li>• Social, economic and political influences shaping methods</li> <li>• Sustainable strategies and evaluation; conclusion</li> </ul> <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 306-337; AQA spec 3.5.1.</p> | 25    |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 11.2     | <p><b>Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable.</b></p> <p><b>Levels of response</b></p> <p><b>Level 5 (21-25 marks):</b> Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.</p> <p><b>Level 4 (16-20 marks):</b> Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.</p> <p><b>Level 3 (11-15 marks):</b> Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.</p> <p><b>Level 2 (6-10 marks):</b> Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.</p> <p><b>Level 1 (1-5 marks):</b> Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (students are not expected to cover all of these; credit other relevant material)</p> <ul style="list-style-type: none"> <li>• Marine productivity: light, nutrients, upwelling; where fisheries occur</li> <li>• Methods: demersal / pelagic trawling, purse seining, drift nets, long lines, shellfish traps - effectiveness, selectivity, energy, impacts</li> <li>• Impacts: overfishing (r vs k selection, catchable vs breeding size), changed age structure, by-catch (albatross, porpoise, turtles, shrimp), ghost fishing, seabed / reef / seagrass damage, dynamite, food web effects (sandeels, tritons)</li> <li>• Population dynamics and MSY: Russell formula, difficulties of data, indicators from catches (CPUE, mean size, age), research on eggs and larvae</li> <li>• Management: quotas, mesh design, escape panels, TEDs, pingers, hook design, effort limits, method bans, NTZs, closed seasons, size limits, protected individuals, population seeding, biodegradable gear; EU CFP, IWC, CCAMLR</li> <li>• Evaluation with examples (cod, orange roughy, herring recovery, NTZ spillover); role of aquaculture; judgement</li> </ul> <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 338-351; AQA spec 3.5.2.</p> | 25    |